

```
CREATE DATABASE ecommerce_db;
```

```
USE ecommerce_db;
```

```
create table users(  
  user_id int primary key,  
  name varchar(50) not null,  
  email varchar(100) unique,  
  manager_id int null  
);
```

```
create table products(  
  product_id int primary key,  
  name varchar(50) not null,  
  price decimal(10,2) not null  
);
```

```
create table orders(  
  order_id int primary key,  
  user_id int,  
  order_date date,  
  foreign key(user_id) references users(user_id)  
);
```

```
show tables;
```

```
create table order_items(  
  order_item_id int primary key,  
  order_id int,  
  product_id int,  
  quantity int,  
  foreign key(order_id) references orders(order_id),
```

foreign key(product_id) references products(product_id)

);

INSERT INTO users(user_id,name,email,manager_id) VALUES

(1,'Ellie','ellie@gmail.com',NULL),

(2,'John','john@gmail.com',1),

(3,'Sarah','sarah@gmail.com',1),

(4,'David','david@gmail.com',2),

(5,'Emma','emma@gmail.com',2),

(6,'Mike','mike@gmail.com',3),

(7,'Sophia','sophia@gmail.com',3);

Output;

```
3 15 19:41:43 INSERT INTO users(user_id,name,email,manager_id)
VALUES (1,'Ellie','ellie@gmail.com',NULL),
(2,'John','john@gmail.com',1),
(3,'Sarah','sarah@gmail.com',1),
(4,'David','david@gmail.com',2),
(5,'Emma','emma@gmail.com',2),
(6,'Mike','mike@gmail.com',3),
(7,'Sophia','sophia@gmail.com',3)
```

7 row(s)
affected
Records: 7
Duplicates: 0
Warnings: 0
0.031 sec

	user_id	name	email	manager_id
▶	1	Ellie	ellie@gmail.com	NULL
	2	John	john@gmail.com	1
	3	Sarah	sarah@gmail.com	1
	4	David	Sarah@gmail.com	2
	5	Emma	emma@gmail.com	2
	6	Mike	mike@gmail.com	3
	7	Sophia	sophia@gmail.com	3
✱	NULL	NULL	NULL	NULL

INSERT INTO products VALUES

(101,'Laptop',70000),

(102,'Mouse',1000),

(103,'Keyboard',2500),

```
(104,'Monitor',15000),
(105,'Headphones',3000),
(106,'Webcam',4500),
(107,'SSD',6000),
(108,'Printer',12000);
```

Output;

```

          INSERT INTO products VALUES (101,'Laptop',70000),
          (102,'Mouse',1000), (103,'Keyboard',2500),
3 17 19:42:57 (104,'Monitor',15000), (105,'Headphones',3000),
          (106,'Webcam',4500), (107,'SSD',6000),
          (108,'Printer',12000)

```

8 row(s) affected
Records: 0.016
8 Duplicates: sec
0 Warnings: 0

Result Grid			
Filter Rows:			
	product_id	name	price
▶	101	Laptop	70000.00
	102	Mouse	1000.00
	103	Keyboard	2500.00
	104	Monitor	15000.00
	105	Headphones	3000.00
	106	Webcam	4500.00
	107	SSD	6000.00
	108	Printer	12000.00
✱	NULL	NULL	NULL

```
INSERT INTO orders VALUES
```

```
(1001,1,'2025-01-10'),
(1002,2,'2025-01-11'),
(1003,1,'2025-01-15'),
(1004,3,'2025-01-20'),
(1005,4,'2025-01-22'),
(1006,2,'2025-01-25');
```

Output;

68 • `select * from orders;`

Result Grid   Filter Rows:

	order_id	user_id	order_date
▶	1001	1	2025-01-10
	1002	2	2025-01-11
	1003	1	2025-01-15
	1004	3	2025-01-20
	1005	4	2025-01-22
	1006	2	2025-01-25
✱	NULL	NULL	NULL

```

INSERT INTO orders VALUES (1001,1,'2025-01-10'), (1002,2,'2025-01-11'), (1003,1,'2025-01-15'),
3 19 19:44:50 (1004,3,'2025-01-20'), (1005,4,'2025-01-22'), (1006,2,'2025-01-25')
6 row(s) affected
Records: 6 Duplicates: 0 Warnings: 0
0.000 sec

```

```
INSERT INTO order_items VALUES
```

```

(1,1001,101,1),
(2,1001,102,2),
(3,1002,104,1),
(4,1003,103,1),
(5,1003,105,2),
(6,1004,102,3),
(7,1004,106,1),
(8,1005,101,1),
(9,1005,107,2),
(10,1006,105,3);

```

Output;

```

INSERT INTO order_items VALUES
(1,1001,101,1), (2,1001,102,2), (3,1002,104,1),
3 21 19:46:22 (4,1003,103,1), (5,1003,105,2), (6,1004,102,3),
(7,1004,106,1), (8,1005,101,1), (9,1005,107,2),
(10,1006,105,3)

```

10 row(s) affected
Records: 0.016
10 Duplicates: sec
0 Warnings: 0

```
SELECT * FROM users;
```

```
SELECT * FROM products;
```

```
SELECT * FROM orders;
```

```
SELECT * FROM order_items;
```

Result Grid				
Filter Rows:				
Edit:				
	order_item_id	order_id	product_id	quantity
▶	1	1001	101	1
	2	1001	102	2
	3	1002	104	1
	4	1003	103	1
	5	1003	105	2
	6	1004	102	3
	7	1004	106	1
	8	1005	101	1
	9	1005	107	2
	10	1006	105	3
✱	NULL	NULL	NULL	NULL

-- cross join - Combine every row from first table with every row from second table

```
SELECT
```

```
u.name,
```

```
p.name AS product_name
```

```
FROM users u
```

```
CROSS JOIN products p;
```

84 • `select u.name,p.name as product_name from users u cross join products p;`

	name	product_name
▶	Sophia	Laptop
	Mike	Laptop
	Emma	Laptop
	David	Laptop
	Sarah	Laptop
	John	Laptop
	Ellie	Laptop
	Sophia	Mouse
	Mike	Mouse
	Emma	Mouse
	David	Mouse

-- A customer called asking about their order.

-- We need to know which customer placed which order.

-- Customer names are in customers table and orders are in diff table so we join both

SELECT

u.name,

o.order_id

FROM users u

INNER JOIN orders o

ON u.user_id=o.user_id;

```

86 • select u.name,o.order_id,o.order_date from users u
87 join orders o on u.user_id = o.user_id;

```

	name	order_id	order_date
▶	Ellie	1001	2025-01-10
	John	1002	2025-01-11
	Ellie	1003	2025-01-15
	Sarah	1004	2025-01-20
	David	1005	2025-01-22
	John	1006	2025-01-25

-- Show us the customer name along with the order date."

SELECT

u.name,

o.order_id,

o.order_date

FROM users u

INNER JOIN orders o

ON u.user_id=o.user_id;

```
86 • select u.name,o.order_id,o.order_date from users u
87 join orders o on u.user_id = o.user_id;
```

Result Grid			
Filter Rows:			
Export:  Wrap Cell Content: 			
	name	order_id	order_date
▶	Ellie	1001	2025-01-10
	John	1002	2025-01-11
	Ellie	1003	2025-01-15
	Sarah	1004	2025-01-20
	David	1005	2025-01-22
	John	1006	2025-01-25

-- We have an Order ID, but we need to know which products belong to each order

-- and how many items need to be packed

```
select u.name ,o.order_id,p.product_id,p.name as product_name,oi.quantity from users u
join orders o on u.user_id=o.user_id
join order_items oi on o.order_id=oi.order_id
join products p on p.product_id=oi.product_id
;
```

Output;

```
89 • select u.name ,o.order_id,p.product_id,p.name as product_name,oi.quantity from users u
90 join orders o on u.user_id=o.user_id
91 join order_items oi on o.order_id=oi.order_id
92 join products p on p.product_id=oi.product_id
93 ;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

	name	order_id	product_id	product_name	quantity
▶	Ellie	1001	101	Laptop	1
	Ellie	1001	102	Mouse	2
	Ellie	1003	103	Keyboard	1
	Ellie	1003	105	Headphones	2
	John	1002	104	Monitor	1
	John	1006	105	Headphones	3
	Sarah	1004	102	Mouse	3
	Sarah	1004	106	Webcam	1
	David	1005	101	Laptop	1
	David	1005	107	SSD	2

-- We want a list of all registered customers and their orders,

-- including customers who have never placed an order."

```
select u.user_id,u.name,o.order_id from users u
```

```
left join orders o on u.user_id=o.user_id;
```

Output;

```
96 • select u.user_id,u.name,o.order_id from users u
97 left join orders o on u.user_id=o.user_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Co

	user_id	name	order_id
▶	1	Ellie	1001
	1	Ellie	1003
	2	John	1002
	2	John	1006
	3	Sarah	1004
	4	David	1005
	5	Emma	NULL
	6	Mike	NULL
	7	Sophia	NULL

-- Give me customers who have never placed any order

```
select u.name from users u
```

```
left join orders o on u.user_id = o.user_id
```

```
where order_id is null;
```

Output;

```
99 • select u.name from users u
100 left join orders o on u.user_id = o.user_id
101 where order_id is null;
```

Result Grid	Filter Rows:	Export:	Wrap
name			
Emma			
Mike			
Sophia			

-- Verify that every order has a corresponding customer.

```
SELECT
```

```
u.name,
```

```
o.order_id
```

```
FROM users u
```

```
RIGHT JOIN orders o
```

```
ON u.user_id=o.user_id;
```

Output;

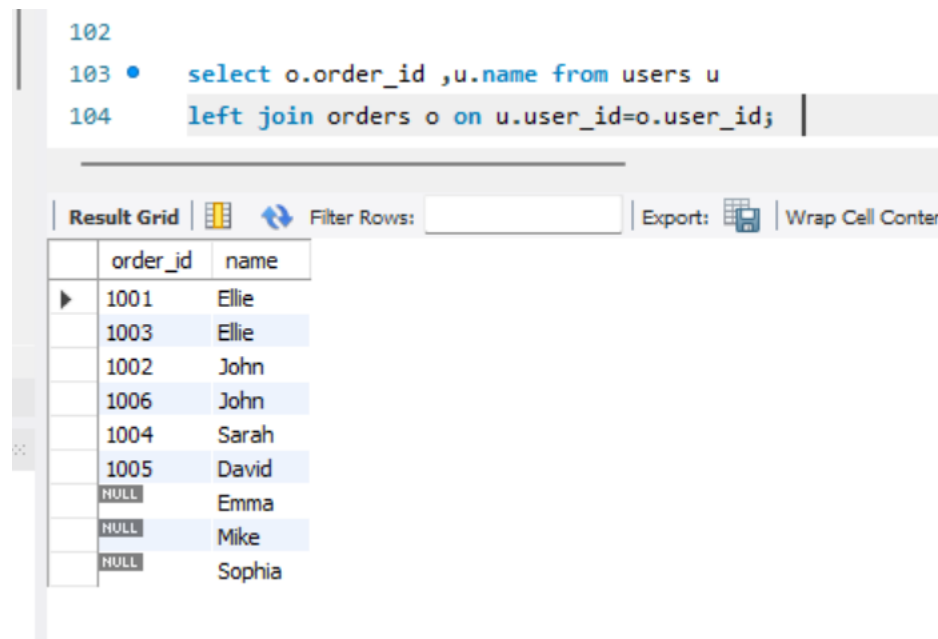
```
103 • select o.order_id ,u.name from users u
104 right join orders o on u.user_id=o.user_id;
```

Result Grid	Filter Rows:	Export:	Wrap Cell
order_id	name		
1001	Ellie		
1003	Ellie		
1002	John		
1006	John		
1004	Sarah		
1005	David		

```
-- Generate a report containing every customer and every order,
-- even if there is no matching record.
```

```
select o.order_id ,u.name from users u
left join orders o on u.user_id=o.user_id;
```

Output;



The screenshot shows a SQL query editor with the following query:

```
102
103 • select o.order_id ,u.name from users u
104 left join orders o on u.user_id=o.user_id;
```

Below the query editor is a 'Result Grid' tab. The grid displays the results of the query:

order_id	name
1001	Ellie
1003	Ellie
1002	John
1006	John
1004	Sarah
1005	David
NULL	Emma
NULL	Mike
NULL	Sophia

```
ALTER TABLE users
```

```
ADD manager_id INT;
```

```
UPDATE users SET manager_id=NULL WHERE user_id=1;
```

```
UPDATE users SET manager_id=1 WHERE user_id=2;
```

```
UPDATE users SET manager_id=1 WHERE user_id=3;
```

```
UPDATE users SET manager_id=2 WHERE user_id=4;
```

```
UPDATE users SET manager_id=2 WHERE user_id=5;
```

Output;

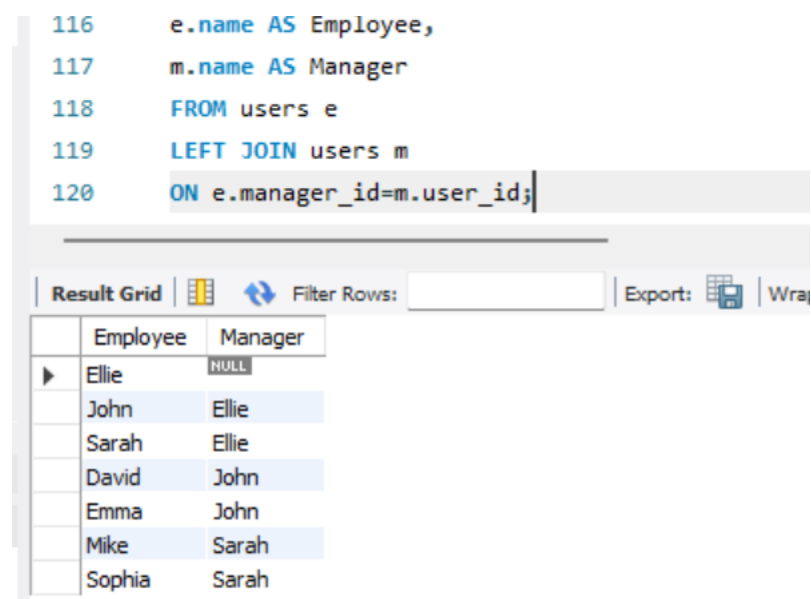
```
3 46 20:33:33 UPDATE users SET manager_id=2 0 row(s) affected Rows matched: 0.000
WHERE user_id=5 1 Changed: 0 Warnings: 0 sec
```

```
-- Display every employee along with their manager
```

```

SELECT
e.name AS Employee,
m.name AS Manager
FROM users e
LEFT JOIN users m
ON e.manager_id=m.user_id;

```



The screenshot shows a SQL query editor with the following query:

```

116 e.name AS Employee,
117 m.name AS Manager
118 FROM users e
119 LEFT JOIN users m
120 ON e.manager_id=m.user_id;

```

Below the query, the results are displayed in a grid. The grid has two columns: 'Employee' and 'Manager'. The results are as follows:

Employee	Manager
Ellie	NULL
John	Ellie
Sarah	Ellie
David	John
Emma	John
Mike	Sarah
Sophia	Sarah

```

-- Total amount spent by each customer

SELECT
u.name,
SUM(p.price * oi.quantity) AS Total_Spent
FROM users u
LEFT JOIN orders o
ON u.user_id=o.user_id
LEFT JOIN order_items oi
ON o.order_id=oi.order_id
LEFT JOIN products p
ON oi.product_id=p.product_id
GROUP BY u.name;

```

```
128     LEFT JOIN order_items oi
129     ON o.order_id=oi.order_id
130     LEFT JOIN products p
131     ON oi.product_id=p.product_id
132     GROUP BY u.name;
```

Result Grid   Filter Rows: Export:  Wrap Cell Content: 

	name	Total_Spent
▶	Ellie	80500.00
	John	24000.00
	Sarah	7500.00
	David	82000.00
	Emma	NULL
	Mike	NULL
	Sophia	NULL